INSYS6112

Assessment 2

20 September 2026

Hazel Kruger
ST10506431@myemeris.edu.za

Question 1

There are servals factors one needs to consider when planning on how to store and manage data. Some of these factors include types of data, how different pieces of data relate to each other, how much data is needed to store and how it must be structured. Based on the needs of the large-scale e-commerce platform might benefit from the use of a NoSQL database.

Also referred to as a 'not only SQL' database. NoSQL databases usually store its data in value key pairs (Butt et al., 2021). Unlike a traditional SQL database, the NoSQL databases store its data in either semi-structured (such as JSON or XML) or even unstructured formats and it also allows for the storage of many different data types such as pictures, documents and video (Butt et al., 2021).

The reason I recommend the NoSQL database is because the e-commerce platform requires the storage of different file / media types. As well the e-commerce platform needs to store its data in both semi-structed and unstructured formats to accommodate how different products need vastly different product attributes. Additionally, NoSQL databases are better at handling high volumes of data as well can provide fast querying and allow for better scalability than a SQL database (Amazon, 2023).

As mentioned earlier, NoSQL databases usualy stores its data in a value key pair (Amazon, 2023). The value of this data can vary depending on the needs of the system. Data can be stored in a semi-structured format such as a JSON or XML file with each file representing a different product with its own unique structure or it can store and link other media such as video and images (Butt et al., 2021). Some types of NoSQL databases may even use tags instead of a key to allow for a broader querying ability (Amazon, 2023).

There are many types of NoSQL databases. Four I would like to discuss are value key databases, document store databases, coloum-oriented databases and graph databases.

Value Key databases are some of the simplest databases. As expected, they store their data in value key pairs with the key acting as a unique identifier for the data (Coronel et al., 2020). They store the data in either text formats or even a document (Coronel et al., 2020). A major benefit of these types of databases is that they allow for fast creation, updating, deletion and fetching of data however, they do not allow for the querying of data (Coronel et al., 2020). As a result, I would not recommend a value key database for the e-commerce platform.

Some ecamples of value key databases include: Dynamo, Riak and Voldemort (Coronel et al., 2020)

Column-oriented databases are very familiar to the traditional SQL databases (Coronel et al., 2020). They differ in that they store their data in a column orientated manner as opposed to a row orientated manner (Coronel et al., 2020). This allows for faster querying of data as only the need columns are fetched. However, column-oriented databases still require structured data models (Coronel et al., 2020) so they are also not recommended for the e-commerce website. Some examples of column-oriented databases include: HBase, Hypertable and Cassandra (Coronel et al., 2020).

Document store databases are like value key databases but instead of value key pairs, they store data in tags and values and can also store documents and not text (Coronel et al., 2020). The use of tags instead of keys allows for the querying of data (Coronel et al., 2020). Additionally, some document stores even allow for aggregate functions to be done on the data such as addition, counting and averaging (Coronel et al., 2020). Out of the three types of NoSQL databases discussed so far, a document store is the most appropriate for e-commerce platform. Some examples of document store databases include MongoDB, OrientDb and RavenDB (Coronel et al., 2020).

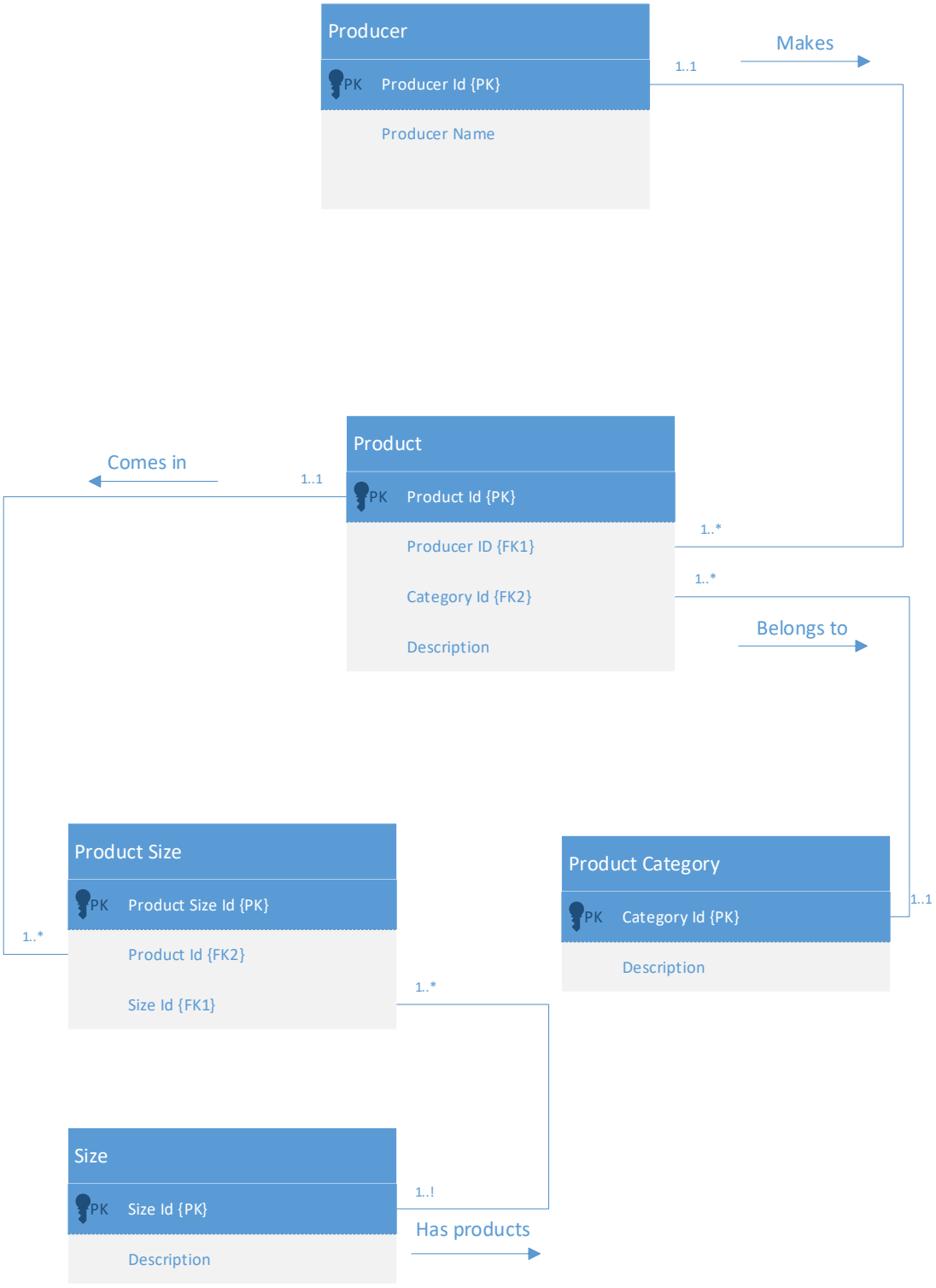
Lastly Graph databases are a type of database that stores its data as nodes. These types of databases are used when the relationship between nodes is considered very important (Coronel et al., 2020). These databases use graph theory and math to link the various nodes to create these relationships (Coronel et al., 2020). Graph databases are often used by social media websites (Coronel et al., 2020 and they also function very well with AI (Coronel et al., 2020). A graph database would also be very appropriate for the e-commerce platform. Some examples include: Neo4j, ArangoDB and GraphBase (Coronel et al., 2020).

As discussed, part of the reason for my recommendation for a NoSQL database is the fact that the e-commerce platform needs to store and process large amounts of data. We can break down this large amount of data (big data) into three categories, volume, velocity and variety (Coronel et al., 2020). Volume is how much data is being stored (Coronel et al., 2020). The e-commerce website for example, is storing data for numerous products that are sold around the world. Variety is how many different types of data need to be stored (Coronel et al., 2020). The online store has many different types of data to be stored such as product catalogues, user reviews shipping information as well as images and videos from sellers. And finally, velocity refers to

the rate at which the data is being collected and proceed (Coronel et al., 2020). The e-commerce site will have a rather high velocity as it is keeping track of clickstream data, search queries as well as social media interactions.

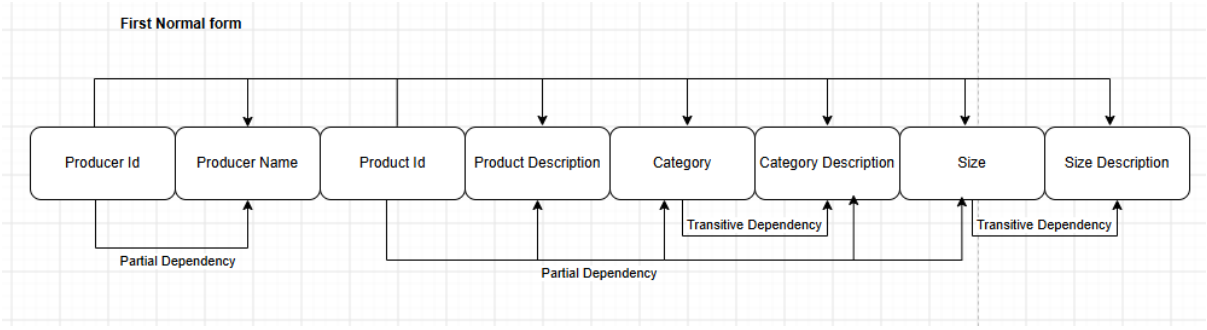
With the need for semi-structed data models, variety in the types and the high volume of data needed to be stored. I believe that a NoSQL database is really the best choice for this online store. As NoSQL databases handle non rigid data models as well as high volumes of data better than their relational database counterparts (Coronel et al., 2020).

Question 2

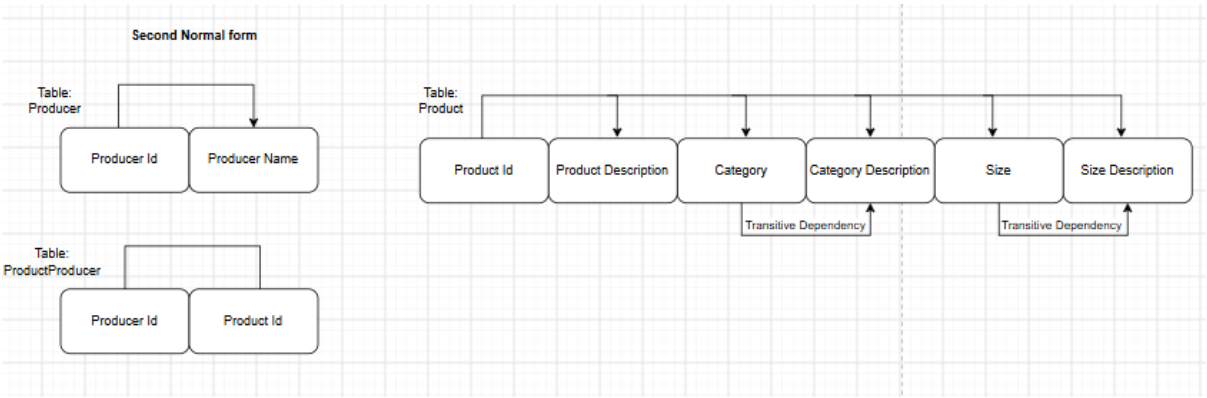

Producer
PK
Producer Id {PK}
Producer Name
Makes
1..1
Product
PK
Product Id {PK}
Producer ID {FK1}
Category Id {FK2}
Description
Comes in
1..1
1..*
1..*
Belongs to
Product Size
PK
Product Size Id {PK}
Product Id {FK2}
Size Id {FK1}
1..*
Product Category
PK
Category Id {PK}
Description
1..1
Size
PK
Size Id {PK}
Description
1..!
Has products

Question 3

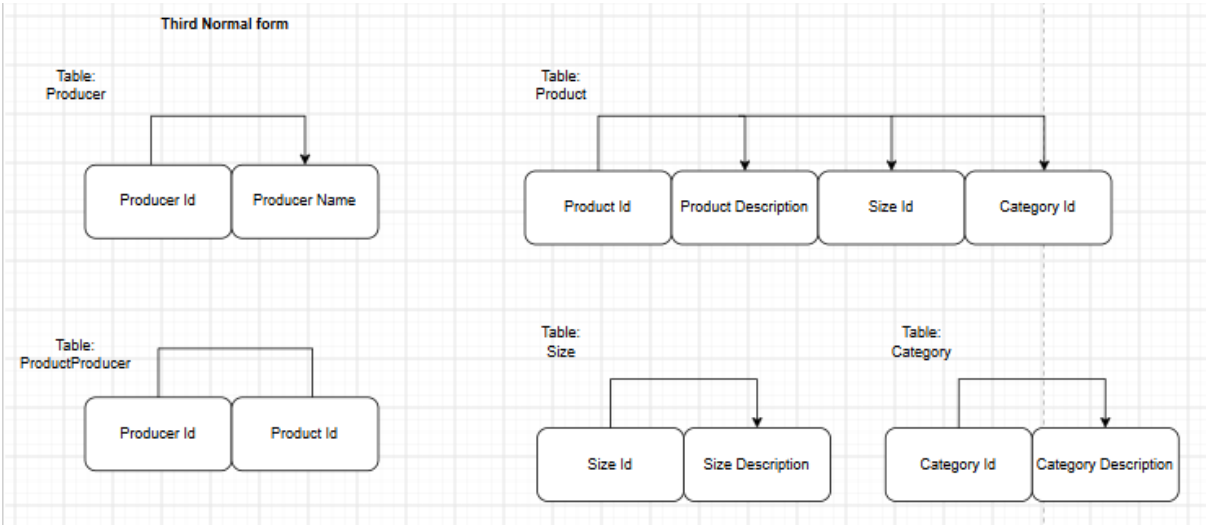
Q.3.1



Q.3.2



Q3.3



Question 4

Q.4.1

```
CREATE SCHEMA Artisan_Cheese;
USE Artisan_Cheese;

create table producer
(
          id int auto_increment not null,
    name varchar(20) not null,
    primary key (id)
);

create table product_category
(
          id int auto_increment not null,
    description varchar(20) not null,
    primary key (id)
);

create table product
(
          id int auto_increment not null,
    description varchar(20) not null,
    producer_id int not null,
    category_id int not null,
    primary key (id),
    foreign key (producer_id) references producer(id),
          foreign key (category_id) references product_category(id)
);

create table size
(
          id int auto_increment not null,
    description varchar(20) not null,
    primary key (id)
);

create table product_size
(
          id int auto_increment not null,
    product_id int not null,
    size_id int not null,
    primary key (id),
    foreign key (product_id) references product(id),
          foreign key (size_id) references size(id)
);
```

Q.4.2

```
insert into producer (name) values ('Clover');
insert into producer (name) values ('Cheese whiz');

insert into product_category (description) values ('Hard');
insert into product_category (description) values ('Soft');
insert into product_category (description) values ('Blue');

insert into product (producer_id, category_id, description) values (2,3,'Royal Blue');
insert into product (producer_id, category_id, description) values (2,2,'Soft Melt');
insert into product (producer_id, category_id, description) values (1,2,'Hard Chedar');
insert into product (producer_id, category_id, description) values (1,3,'White cream');

insert into size (description) values ('250g Wheel');
insert into size (description) values ('500 Wheel');

insert into product_size (product_id, size_id) values (1,1);
```

```
insert into product_size (product_id, size_id) values (1,2);
insert into product_size (product_id, size_id) values (2,1);
insert into product_size (product_id, size_id) values (2,2);
insert into product_size (product_id, size_id) values (3,1);
insert into product_size (product_id, size_id) values (3,2);
insert into product_size (product_id, size_id) values (4,1);
insert into product_size (product_id, size_id) values (4,2);
```

Q.4.3

```
update producer
set name = 'Lovely Cheese'
where id = 1;
```

Q.4.4

```
delete
from product_size
where id = 8;
```

References

Amazon, 2023. [online] Relational vs nonrelational databases - difference between types of databases - AWS. Available at: <https://aws.amazon.com/compare/the-difference-between relational-and-non-relational-databases/> [Accessed 22 Aug. 2026].

Butt, M.H., Siddiqui, A.F. and Haswani, P., 2021. Comprehensive Analysis on SQL and No SQL. *Asian Journal of Engineering, Sciences & Technology (AJEST)*, [online] 11(1), pp.15–24. Available at: <https://research.ebsco.com/plink/df70890b-b7b7-35b3-939d-c2424ce2bbf0> [Accessed 25 Sep. 2026].

Coronel, C., Blewitt, C., Crockett, K. and Morris, S., 2020. Database principles: Fundamentals of design, implementation, and management. 3rd ed. Australia: Cengage.